

Resilience Simulation: What and Why

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Why Resilience?



Engineered systems are “brittle”



Natural systems “bounce back”

What if we could adapt to hazardous events so they don't become catastrophes?

- Disasters and the unpreventable – How do we respond when we can't preempt?
- Autonomy and new technologies—How do we engineer hazard mitigation?

What is Resilience?

A widely agreed-upon definition is:

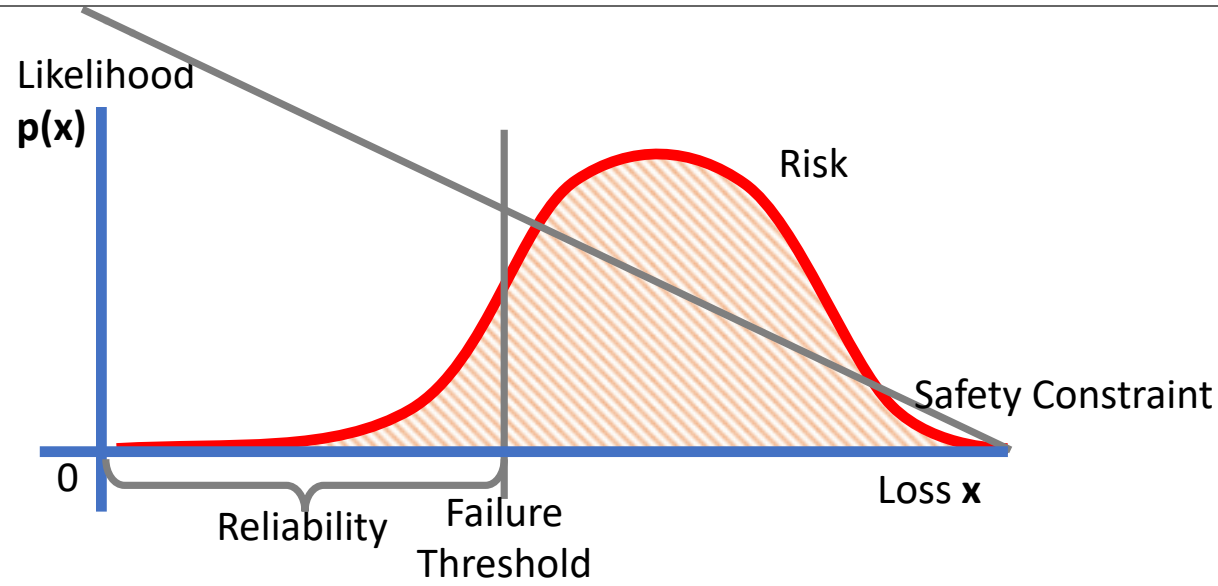
(The) ability to prepare and plan for, absorb, recover from, and more successfully adapt to adverse events

National Academies, Policy, Global Affairs, Committee on Science, Public Policy, & Committee on Increasing National Resilience to Hazards. (2012). Disaster resilience: A national imperative. National Academies Press.

Key Points:

- Resilience is understood as an ability: Adaptive Capacity
 - e.g., the system can prepare for and/or mitigate given condition(s) A system can be:
- Resilient to X (specified condition, hazard, scenario, disruption or change etc.)
- Generically resilient—to large numbers of scenarios, unspecified or unknown conditions

Related Concepts You May Have Heard of



Risk: The effect of uncertainty on objectives (ISO 31073:2022)

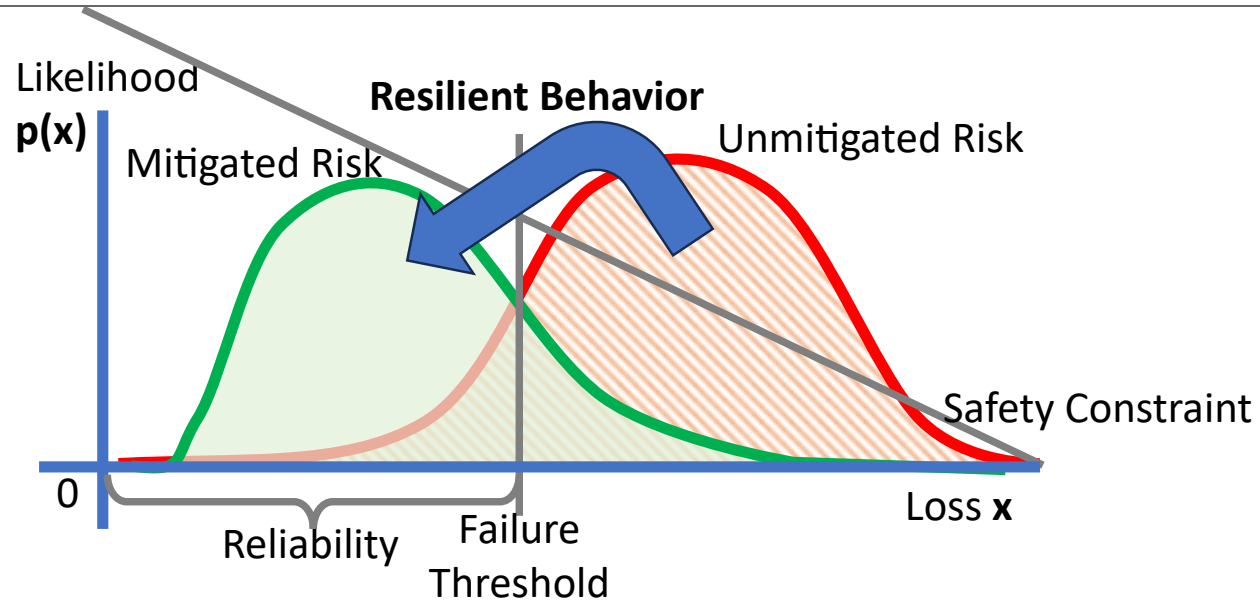
$$Risk = \int p(x) * x dx$$

Reliability: “the probability of an item performing its purpose adequately for the period of time intended under the operating and environmental conditions encountered” (KK Aggarwal, 1993)

$$Reliability = \int_0^{x_f} p(x) dx = 1 - \int_{x_f}^{\infty} p(x) dx$$

Safety: “freedom from risk which is not tolerable” (ISO Guide 51)

Connection of Resilience to Resilience to Risk and Safety



Resilience is a **behavior the system performs** to mitigate risk, e.g., recovering performance, switching to a redundant system, or reconfiguring the mission

One result of this is that risk mitigation is a good (but not the sole) measure of resilience:

Unmitigated Risk $R_u = \int p_u(x) * x \, dx$

Mitigated Risk $R_m = \int p_m(x) * x \, dx$

Risk Mitigation $\Delta R = R_u - R_m$

What's different about resilience?

Resilience is about the dynamics, interactions, and control of hazardous scenarios.

- Traditional risk and safety processes are generally about **quantifying probability** of “success” versus “failure” and may rely on relatively **simple models** to do so
- Resilience is about navigating the **complex interactions** between system behaviors to improve hazard mitigation via a **variety of metrics** (e.g., recovery, robustness, etc.)

Again, risk, reliability, safety, and resilience are **not mutually exclusive concepts**—the main difference is what the given frameworks emphasize

How to think about resilience simulations

A simulation tells us how resilient a system is or can be with a given set of features

All models are wrong, some are useful

-George Box

Simulation of resilience can be thought of as a game with: - The behaviors the system is going to embody and how they interact - The conditions the system is going to operate over - What is the system's strategy for dealing with these scenarios? - What are the good or bad outcomes - We can call the system more resilient when those strategies improved the outcome(s) There may be many metrics to capture this this!

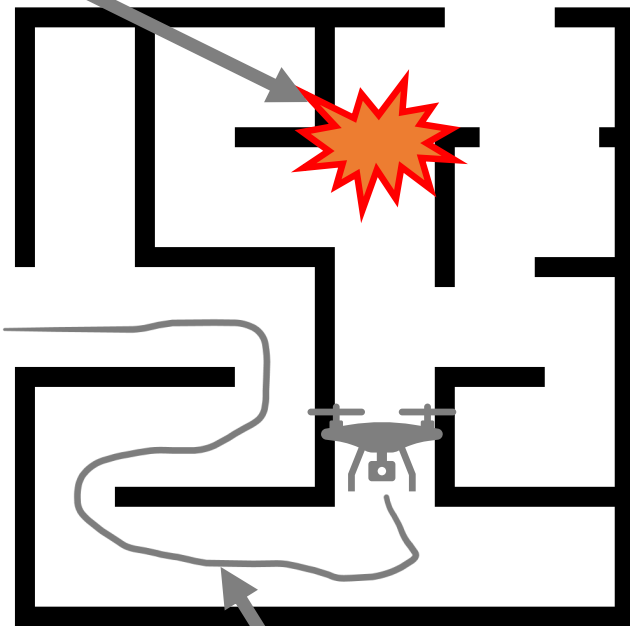
What does that look like?

Scenario Setup

Condition: Hazard blocks path

Finish

Start



Behavior: Locomotion and navigation through airspace

Resilience Simulation

Outcome:
Crash 😞

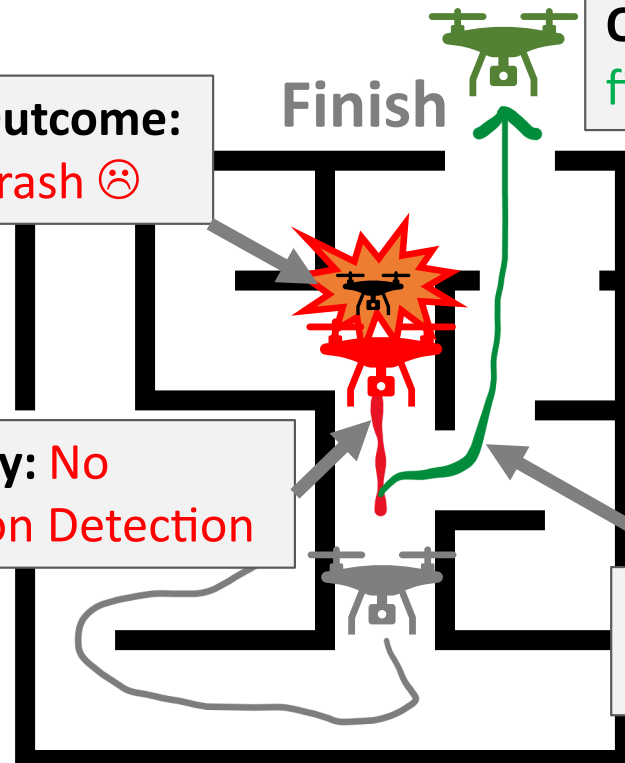
Finish

Outcome: Maze finished! 😊

Start

Strategy: No
Intrusion Detection

Strategy: Detect
and re-plan!



Things to consider:

- Our model can be pretty abstract and still give us answers we can use
- There may be multiple places the drone could be blocked!

- If we really want to know how well this strategy performs overall, we may want to test it in a wide range of scenarios

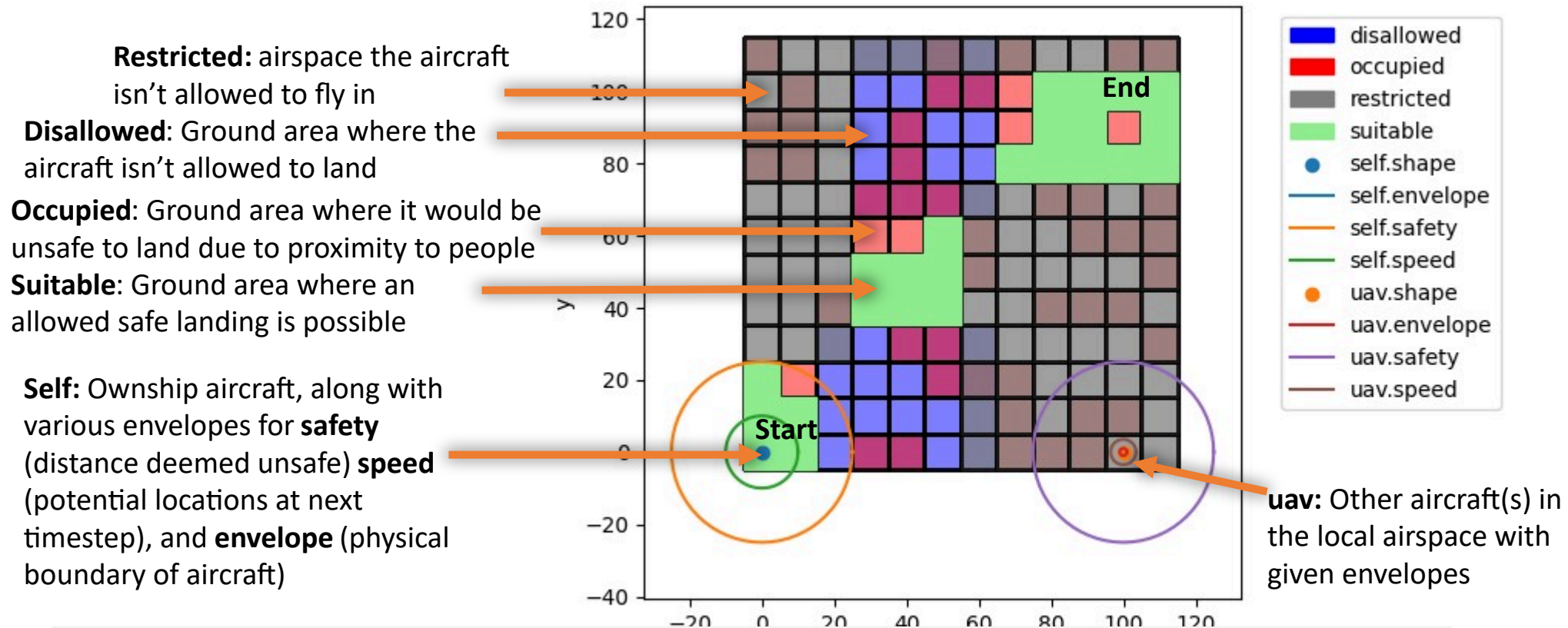
How does fmdtools help?

- Representing the complexity required for resilience analysis: by providing flexible, composable modeling paradigm
- Reducing setup costs: By providing built-in analysis and simulation methods
- Providing capabilities aligned with our research on (1) sampling the hazardous state-space, (2) representing human agents in these systems and (3) optimizing model resilience metrics

Demonstrative Example: Drone Proximity to Threat Evaluation

- Created adaptable domain-specific library that can be used to model aircraft and drones in the airspace
- Focused on safety to collision with other aircraft as well as landing
- Question “How does state awareness of other air threats improve drone resilience to collision?”
- Model at [examples/airspacelib/contingency_management](#)

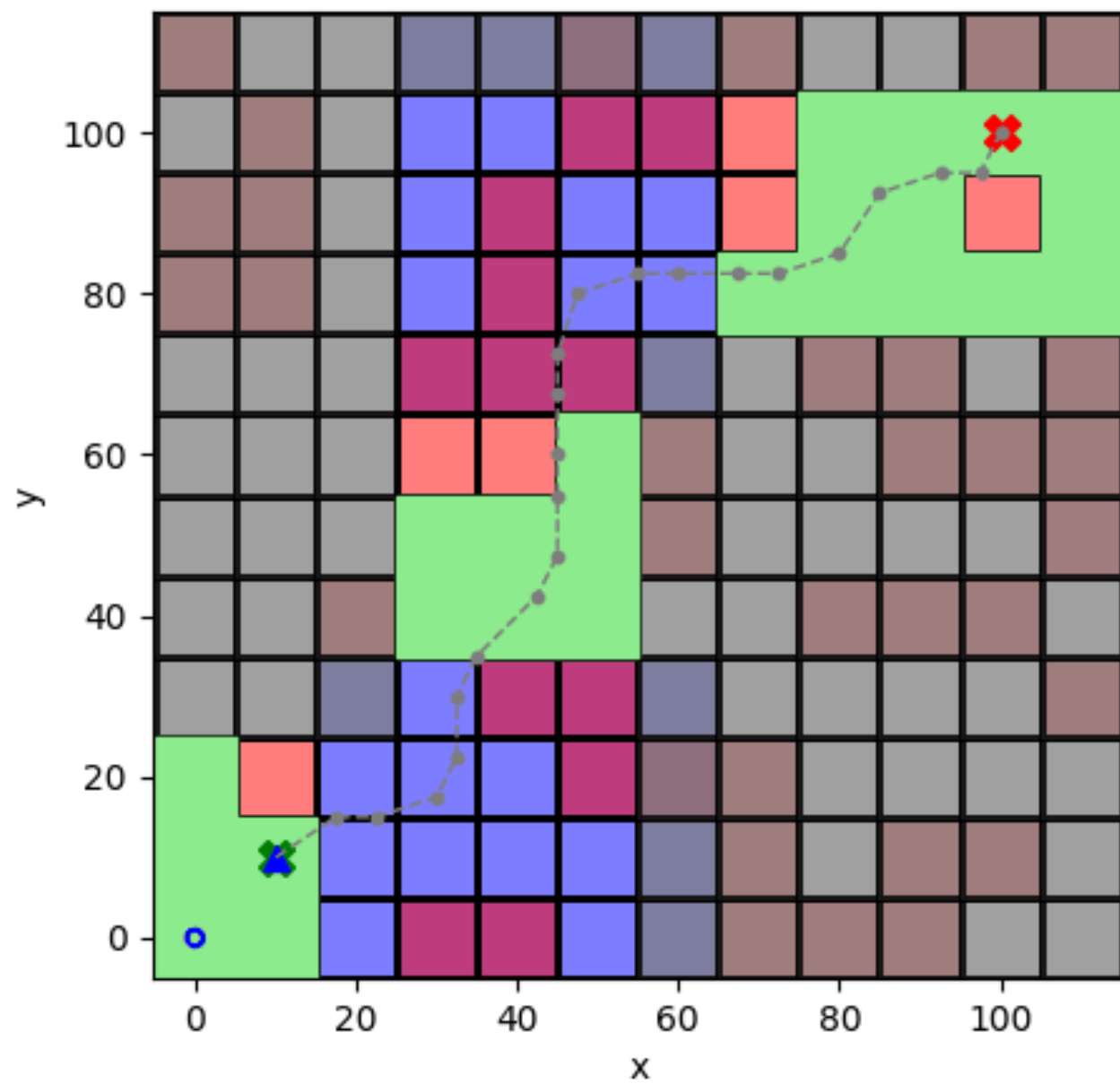
Model Setup



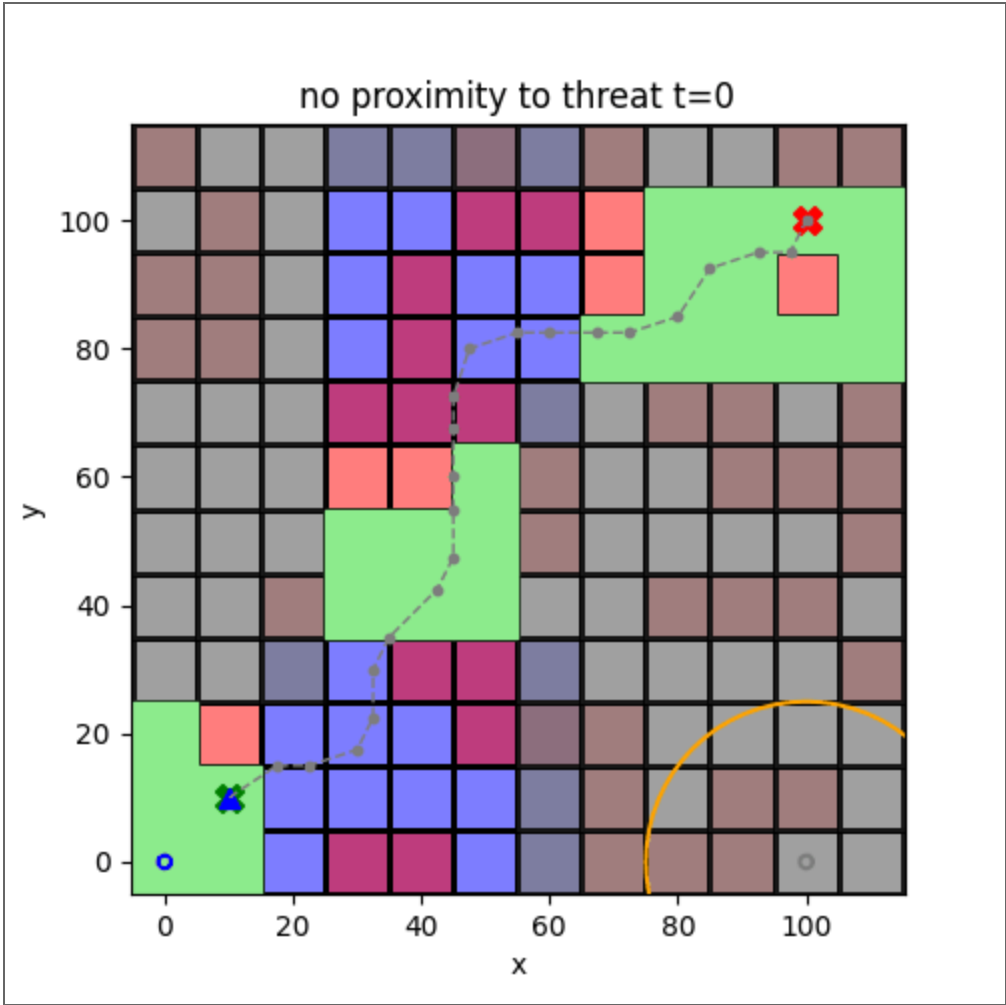
Aspects of the map are randomly generated in this instance— but parameterizable such that we can change locations (start, end) and distributions (e.g., to match a real environment for a mission).

Simulation - Nominal

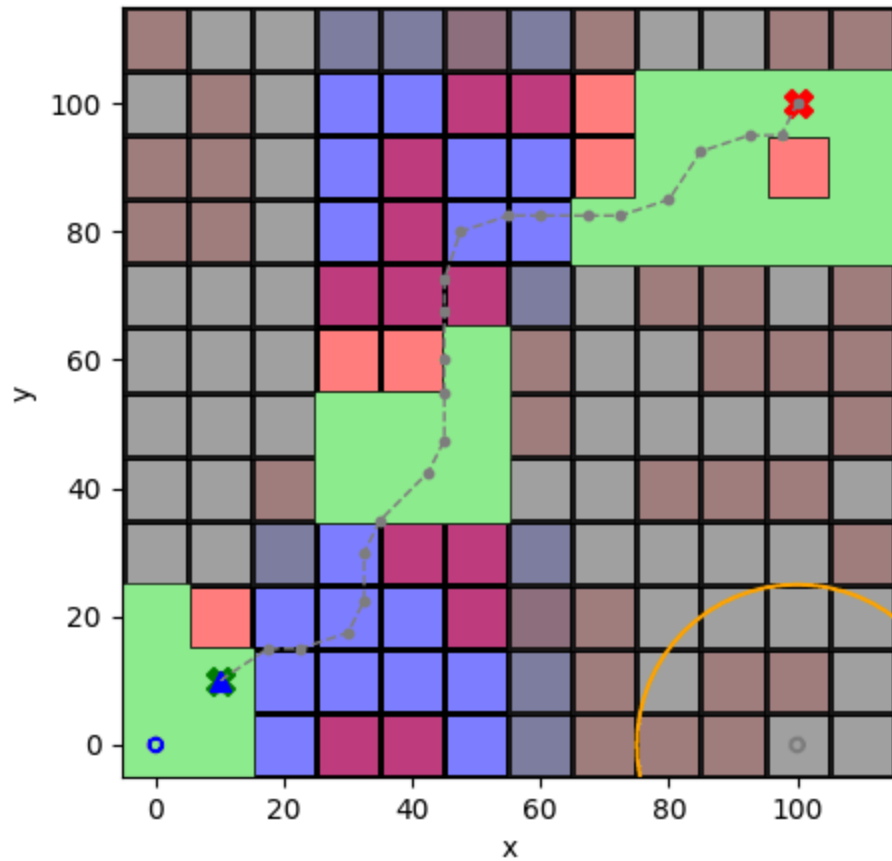
t=0



Simulation - Hazards



with proximity to threat $t=0$



- Without proximity to threat functionality, drone may fly into errant intruding drone
- Proximity to threat functionality causes a pause in mission as well as mission re-planning

Conclusion

- Resilience is the capability of a system to actively prevent, mitigate and/or adapt to hazardous and/or anomalous conditions. To evaluate system resilience, we need ways of capturing:
 - How a system's dynamical behavior(s) intensify or mitigate a given set of hazardous conditions
 - How the system is controlled at a high level, including interactions between its operator/autonomy, physical behaviors, and environment
- Modeling and Simulation gives us the ability to analyze systems resilience to hazardous scenarios
 - Can instantiate wide ranges of scenarios we practically cannot test in the real world or even a more detailed sim
- The fmdtools framework can the foundation for analyzing the resilience of any system of interest

